AI-Assisted Tennis Coaching for Beginners

Video-Based Feedback and Adaptive Practice Planning

1 Overview and objectives

A beginner can book a court and watch a demonstration, yet still struggle to understand why their own forehand is inconsistent or what to practise next. Frequent coaching may be costly: a coach quoted our team more than S$100 per hour. Two one-hour sessions each week would exceed S$800 over four weeks—more than a quarter of the S$3,000 monthly base stipend for international PhD students on the NTU Research Scholarship (NTU, 2026). Court access, time and practice partners also constrain training.

We propose an AI assistant that turns ordinary phone video into focused, evidence-linked feedback and a practical weekly plan. Pose estimation extracts movement features; a pretrained language model uses these features and retrieved coaching references to explain a priority issue. AI supports practice between coaching sessions, with uncertainty made visible and coach review used to check advice.

Our objectives are to build an end-to-end forehand prototype, evaluate whether its feedback is supported by the footage and understandable to beginners, and test whether practice plans respond to users' time, court access and session feedback.

2 Data collection and project scope

Scope. The prototype covers the forehand only, for adult beginners below NTRP 3.0. At this level a forehand has roughly five to eight recurring faults, a bounded set we can evaluate within the semester. Technique material from Tennis Australia (2010) defines what counts as a fault, while its tactical cues (e.g., "clear the net by over a metre") shape how feedback is worded: as an instruction about the ball, not a joint angle. External-focus instructions of this kind improve performance and learning (Chua et al., 2021), including for tennis groundstrokes (Tapan et al., 2023).

Available data. The project materials provide 9 MP4 clips (3 each for forehand, backhand and serve), 2 GIFs, 2 JPGs, 15 example player prompts and the Tennis Australia brochure. The clips show professional players, so we use them as reference examples rather than test cases for beginner faults. Public datasets such as THETIS (Gourgari et al., 2013), which holds labelled forehands from 55 amateur and experienced players, serve only as a pose-estimation benchmark because they were recorded indoors rather than on court.

New recordings. We will record 10–20 clips with consenting adult beginners. Poor repetitions are kept alongside clean ones so the system sees the faults it must detect, and fed balls are labelled separately from rally balls because practice unlike real play transfers poorly (Krause et al., 2018). The camera angle is chosen to match the target faults.

User input. Beginners usually film on a phone held by a friend, so we accept everyday clips: handheld or fixed, any orientation or view and a normal frame rate, as long as one forehand is visible. The player adds their level, dominant hand, goal, question and weekly court access. Tips such as keeping the full body in frame are optional; testing on rough clips shows whether the model works on real beginner footage.

Quality handling. Rough clips are kept, not rejected. Our team labels the stroke and its preparation, contact and follow-through phases, trims frames where the swing is hidden, crops out other people and overlays, and records the camera view and conditions (e.g., handheld, low light). A clip is set aside only if the swing cannot be seen. Each labelled clip and its player record then go to pose estimation.

Ethics. Adults only, with written consent, anonymised filenames, faces obscured in presented material and raw footage deleted after the course. Supplied clips are used only within this project.

3 Video processing, pose estimation and planned tools

The data-processing stage will convert the raw inputs into representations that the AI reasoning module can use. Since model performance is bounded by data quality and representation, this stage focuses on three inputs: athlete video, wearable sensor data, and a coaching knowledge base. The initial prototype is limited to the forehand, so every step below is designed around forehand clips.

For athlete video, each MP4 clip will be decoded into frames and checked for usable full-body visibility. Frames will be resized and trimmed to the stroke so that the player appears at a similar scale across clips. Pose estimation will then extract body keypoints such as the shoulders, elbows, wrists, hips, knees, and ankles. Low-confidence keypoints will be flagged, short gaps will be filled by interpolation, and light temporal smoothing will reduce frame-to-frame jitter. Coordinates will be normalised by body size, such as torso length, so that clips recorded at different distances can be compared more fairly.

Pose classification will act as a check at the start of the pipeline. The pose sequence will be used to label the clip as forehand or not forehand and report a confidence score (for example, forehand at 90%). Clips that are not classified as forehand will be flagged and not passed on for feedback. The 3 backhand and 3 serve clips will serve as negative examples to test this check.

For each forehand clip, we will compute features that match common beginner forehand problems: shoulder-hip rotation, elbow angle at contact, knee bend, trunk lean, wrist position relative to the hip, and the finishing position of the wrist after the swing. Wrist speed will be used to divide the stroke into preparation, contact, and follow-through. Contact will be approximated by the peak in wrist speed, since the ball cannot be tracked reliably from ordinary video.

Wearable sensor data, such as accelerometer, gyroscope, and heart rate readings, will be converted into tabular form with a timestamp, sensor type, value, unit, and body site for each reading. Values will be aligned with the video by time so that a reading, such as wrist gyroscope output, can be linked to a forehand phase. The supplied clips do not include sensor data, so the first prototype will test only the table format and the alignment step.

The knowledge base will contain our coaching references, namely tennis.docx and the Tennis Australia coaching brochure. The text will be split into short chunks, and each chunk will be tagged with its source. For a forehand question and a detected issue, such as “Why is my forehand unstable?”, the system will use retrieval-augmented generation (RAG) to rank the chunks by relevance and return the top results with their sources. The ranked texts will ground the advice in the references and will also serve as the evidence links shown to the user.

The output of this stage will be the phase-labelled forehand features with confidence scores, the stroke label, any sensor table, and the ranked reference texts. All of these will be sent to the AI reasoning module.

The planned tools are Python with OpenCV for video handling, MediaPipe Pose (BlazePose) for pose estimation, and NumPy and pandas for feature extraction, smoothing, and sensor tables. For retrieval, we plan to use sentence-transformers for text embeddings and FAISS for ranking. If MediaPipe performs poorly on our clips, we will test an alternative such as RTMPose. The 3 forehand clips will be the main test of the pipeline, and the other clips, GIFs, and JPGs will be used to check the stroke classification step. We will check that the pipeline runs end to end and that the phase detection is reasonable.

4 AI reasoning, prioritisation and limitations

The AI reasoning module will interpret the movement features and coaching references provided by the data-processing stage. For the initial forehand prototype, it will compare the player’s movement across key phases, such as preparation, contact, and follow-through, with relevant reference information. From these comparisons, the system will identify possible technical issues, which may include timing, balance, body rotation, or follow-through problems.

Because multiple deviations may appear in the same clip, the system will prioritise issues rather than report every detected difference. Prioritisation will consider factors such as the magnitude of the deviation, confidence of the extracted features, consistency across relevant frames, and coaching relevance for a beginner. The reasoning module will output the highest-priority issue together with supporting evidence and a brief explanation. Detailed feedback presentation and practice planning will be handled in the subsequent user-output stage.

The main limitations include errors in pose and motion estimation, sensitivity to camera viewpoint and occlusion, difficulty in accurately estimating racket–ball interaction from ordinary video, variation in valid forehand techniques, and limited generalisation from a small prototype dataset.

5 User feedback and adaptive practice planning

The initial prototype will provide beginner tennis players with a video feedback report focused on the forehand. Each report will highlight one or two priority technique issues, explain them in plain language, and link them to relevant timestamps or annotated frames. Suggested corrections will be supported by coaching references and relevant demonstration clips where available. If the recording is unclear or an issue cannot be assessed reliably, the report will explain the limitation and recommend how to capture a better video.

A personalised weekly practice plan will translate the feedback into specific drills, suggested session durations, and observable practice goals. The plan will consider the player's available time, court access, and whether a practice partner is available. After each session, users will report whether they completed the drills, how difficult they found them, and whether the feedback was helpful. Together with new video observations, this information will help the system adjust the next plan—for example, by repeating a drill, simplifying it, or progressing to another technique issue.

Output evaluation

We propose evaluating the prototype through coach review and a small pilot involving approximately 5–10 beginners. A qualified tennis coach will assess whether the identified issues are supported by the footage, whether their prioritisation is appropriate, and whether the recommended drills address them. Beginners will rate the clarity and usefulness of the report and explain what they would practise next. We will also check whether plan adjustments respond appropriately to session feedback. Where repeat recordings are available, a coach will assess changes using a consistent forehand technique rubric. The pilot will evaluate feasibility and provide preliminary observations of improvement; it will not establish long-term coaching effectiveness.

The 2–4-person pilot in the timeline is an initial usability check within the wider target of 5–10 participants. Findings available by 25 October will inform the presentation; any unmet recruitment target or incomplete follow-up will be reported.

6 Brief literature review

Bazarevsky et al. (2020) proposed BlazePose, a lightweight pose estimation model that predicts 33 body keypoints and runs in real time on mobile devices. This work relates to our video processing section because it supports using a fast, accessible pose estimator on ordinary phone video, which matches how beginners would record their practice.

Yeh et al. (2025) proposed CoachMe, a reference-based sports coaching model that compares a learner's motion with reference movements in temporal and physical dimensions to identify movement errors and generate corrective guidance. This work relates to our AI reasoning section because it supports using reference-based movement comparison to identify and prioritise important technical issues in a player's forehand.

Recent reviews suggest that video-based visual and other augmented feedback can support motor learning, although results depend on the task and feedback design (Mödinger et al., 2022; Moinuddin et al., 2021; Petancevski et al., 2022). Tennis studies on self-controlled video feedback and AI-generated action guidance provide further motivation for combining video evidence with practical coaching suggestions (van der Meer et al., 2024; Wang et al., 2025). These findings support our proposed feedback and practice cycle, but do not directly establish the effectiveness of AI-generated forehand feedback or adaptive weekly plans for beginners.

Together, these studies motivate an accessible pipeline combining pose estimation, reference-based reasoning and actionable feedback. The project will test this combination for beginner forehands; its effectiveness cannot be assumed from the component studies alone.

7 Roles and responsibilities

Member(s)	Responsibility and deliverable
Shao Zirui	Overview, problem definition and objectives; integration and final submission; pilot with Shule.
Cao Hao and Huang Liangcai	Data collection, consent, annotation and quality checks.
Joshua Adrian Cahyono	Video processing, pose estimation and retrieval; structured features and reference passages.
Huang Helong	AI reasoning, issue prioritisation and limitations; evidence-supported issue diagnosis.
Zhao Shule	Feedback reports, adaptive plans and evaluation; pilot with Zirui.
All members	Literature review, testing and presentation; review the integrated draft and seek tutor feedback.

8 Project timeline

Date (2026)	Task	Lead
4–5 Oct	Agree target faults; fix camera angle	All
6–9 Oct	Combine, review and submit proposal	Zirui
10–18 Oct	Recruitment and consent; recording; quality screening	Cao Hao, Liangcai
14–20 Oct	Pose estimation and reasoning on sample clips	Joshua, Helong
21–24 Oct	Pilot with 2–4 beginners; collect user feedback	Shule, Zirui
21–25 Oct	Slides and demo preparation; analyse and integrate pilot findings	All
26 Oct	Group presentation	All

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